MIL-P-23957A(OS)
18 September 1969
SUPERSEDING
MIL-P-23957(Wep)
31 January 1964

MILITARY SPECIFICATION

POWER DRIVE, TRAIN, MK 61 MOD 2 POWER DRIVE, ELEVATION, MK 62 MOD 2 CHASSIS, AMPLIFIER, LD 272613

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - This specification establishes the requirements for the procurement of Power Drive, Train, Mk 61 Mod 2, LD 272638; Power Drive, Elevation, Mk 62 Mod 2, LD 272639 and Chassis, Amplifier, LD 272613. The power drives and associated amplifier chassis position Launcher Mk 112 in train and elevation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-17111	Fluid, Power Transmission
MIL-L-23956A(OS)	Launching Group MK 16 Mod 4

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-419	Cleaning and Protecting Integral Piping for Oil Hydraulic Power Transmission Equipment

FSC 1055

Source: https://assist.dla.mil -- Downloaded: 20260-19T18:29Z
Check the source to verify that this is the current version before use.

DRAWINGS

Naval Ordnance Systems Command

LD 272569	Dummy Director Mk 3 Mod 8 and Error Recorder Mk 12
LD 272600	Receiver-Regulator, Elevation, Mk 47 Mod 2
LD 272601	Receiver-Regulator Train, Mk 46 Mod 2
LD 272613	Chassis, Amplifier
LD 272632	Motor, Electric (Elevation)
LD 272633	Pump and Stroke Control, Hydraulic (Train)
LD 272634	Pump and Stroke Control, Hydraulic (Elevation)
LD 272635	Power Unit (Elevation)
LD 272636	Power Unit (Train)
LD 272637	Tubing Hydraulic (Carriage) (Power Drive Part)
LD 272638	Power Drive, Train Mk 61 Mod 2
LD 272639	Power Drive, Elevation, Mk 62 Mod 2
LD 272643	Test Stand, Dynamic, Elevation Power Drive, Mk 62 Mod 2
LD 272644	Test Stand, Dynamic, Train Power Drive, Mk 61 Mod 2
LD 286717	Motor, Electric (Train)
1284008	Body, Valve Assembly
1284023	Servo-Valve
1478765	Test Panel, Assembly, Amplifier
1676894	Piston
1676907	Piston, Control

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

DRAWINGS

Naval Ordnance Systems Command (Continued)

1683740	Spool, Valve
1683770	Spring
1691753	Motor, Hydraulic, B-End (Train and Elevation)
1778642	Spool, Valve
1778662	Piston
1877515	Amplifier, Train or Elevation, Test Data
2319715	Power Unit Subassembly
2319716	Power Unit Subassembly
Sketch 308777	Alternate Method of Cleaning and Protecting Steel Tubing and Fittings

PUBLICATIONS

Naval Ordnance Systems Command

OD 14430 (Second Revision)	Factory Acceptance Test, Power Drive, Train Mk 61 Mod 2; Power Drive Elevation, Mk 62 Mod 2; Amplifier LD 272613
OD 18388	Preservation, Packing, Shipping, Handling and Storage Instructions
OP 2385, Vol. 2	Launching Group Mk 16 Train Power Drive Mk 61 Elevation Power Drive Mk 62 and Amplifier Mk 153 Description and Maintenance

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

National Bureau of Standards Publications

Handbook H28 and all Supplements thereto
Screw Thread Standards
for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description - The Power Drives and Amplifier described in this document are designed to provide the motive power, under order signals from Fire Control, to position the Launcher Mk 112 Mod 2 in train and elevation within the specified accuracies. They consist of the following major assemblies. All power drive components are mounted on the upper base ring of the launcher. The amplifier is mounted off-launcher.

Power Drive, Train, Mk 61 Mod 2 - LD 272638

Power Unit - LD 272636
Motor, Hydraulic B-End - Drawing 1691753
Receiver-Regulator, Train, Mk 46 Mod 2 - LD 272601
Tubing, Hydraulic - LD 272637 (Power Drive Part)

Power Drive, Elevation, Mk 62 Mod 2 - LD 272639

Power Unit - LD 272635
Motor, Hydraulic B-End - Drawing 1691753
Receiver-Regulator, Elevation, Mk 47 Mod 2 - LD 272600
Tubing, Hydraulic - LD 272637 (Power Drive Part)

Chassis, Amplifier (2) - LD 272613

3.2 General requirements -

3.2.1 Materials - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the Power Drives, Train, Mk 61 Mod 2, Elevation, Mk 62 Mod 2, and Amplifier, Power Drive, shall conform strictly to the detail requirements indicated on the drawings and specifications listed in LD's 272638, 272639 and 272613.

3.2.2 Finishes, coatings, construction and dimensions - Finishes, coatings, construction and dimensions for the components of the Power Drives, Train, Mk 61 Mod 2 and Elevation, Mk 62 Mod 2, and the Amplifier, Power Drive,

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

shall be strictly in accordance with the applicable drawings, specifications and publications listed in LD's 272638, 272639 and 272613.

3.2.3 Moisture, fungus and corrosion - The Power Drives, Train, Mk 61 Mod 2 and Elevation, Mk 62 Mod 2, and the Amplifier, Power Drive shall be rendered moisture, fungus and corrosion resistant in accordance with the detail requirements specified on the drawings, specifications and publications in LD's 272638, 272639 and 272613.

3.2.4 Marking - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in LD's 272638, 272639 and 272613.

3.2.4.1 Name plate weight - The name plate weight is the aggregate of the scale weights of all the parts shown which appear in the list of drawings of the components. Each name plate shall carry the name plate weight of the components to which it is affixed.

3.2.5 Interchangeability - Component parts of the assembled units (unless otherwise specified in the applicable drawings), fabricated in strict accordance with their applicable drawings, shall be interchangeable and shall assemble with mating components of the unit without selection or fitting.

If, during assembly of the fabricated parts of a unit, selection or fitting is required, then changes in dimensions are necessary and revision and correction of the applicable drawings are required. See 3.2.9.

3.2.6 Gages - The supplier shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of the pertinent Navy Final Inspection Gages for guidance in the design of the supplier's inspection gages. However, such procedure shall not serve to relieve the supplier of his responsibility in the design and manufacture of such gages as may be required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and all supplements thereto. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the Supplement to the aforementioned Handbook H28.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

3.2.8 Requests for deviations - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein shall be submitted to the procuring activity. The deviation shall have been authorized by the Naval Ordnance Systems Command or by a Naval activity delegated such authority, prior to the production of the affected item or prior to the performance of a particular operation on the affected item. The request shall define the suggested deviation in sufficient detail to permit the procuring activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated.

3.2.9 Conflicting requirements - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein, shall be referred, in writing, to the procuring activity or appointed agent for interpretation, clarification and resolution (or correction).

3.3 Electrical and hydraulic requirements -

3.3.1 Dielectric strength and insulation resistance - The requirements for dielectric strength and insulation resistance to which this specification is applicable and unless otherwise specified in the individual requirements shall meet the following: Shall withstand a potential of 1,000 volts alternating current (a.c.) sinusoidal for a period of one minute from electrical terminals to case or frame. Shall also show a minimum of three megohms resistance at 500 volts direct current (d.c.) from electrical terminals to case or frame after meeting the requirements of the high potential.

3.3.2 Hydraulic fluid and hydraulic fluid temperature - The requirements for hydraulic fluid and hydraulic fluid temperature to which this specification is applicable, and unless otherwise specified in the individual requirement, shall meet the following: Hydraulic fluid shall conform to MIL-F-17111. Performance characteristics are those obtained with the hydraulic fluid temperature in the range of 90 degrees to 120 degrees Fahrenheit (F).

3.4 Performance requirements and product characteristics - The Power Drive, Train, Mk 61 Mod 2 and Elevation, Mk 62 Mod 2, and Amplifier, Power Drive, shall meet the following performance requirements and product characteristics.

3.4.1 Power unit - The requirements for the power unit assembly, listed as follows, shall be as specified on the drawings of LD's 272636 and 272635.

3.4.1.1 Assembly requirements - The watertightness, hydrostatic and gearing backlash shall be as specified on Drawings 2319716 and 2319715.

3.4.1.2 Hydraulic motor and transfer valve - The hydraulic motor and transfer valve shall meet the requirements of Drawing 1691753.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

3.4.1.3 Hydraulic pump and stroke control, LD's 272633, 272634 - The hydraulic pump and stroke control shall meet the following requirements.

3.4.1.3.1 Internal hydrostatic pressure - The hydraulic pump and stroke control, with the exception of the pump case and stroke control servo-valve housing, shall withstand an internal hydrostatic pressure of 4,500 ±50 pounds per square inch (psi) for a period of ten minutes without showing any signs of leakage, or permanent deformation during or after the test.

The hydraulic pump case and stroke control servo-valve housing shall withstand a hydrostatic pressure of 5 psi for a period of ten minutes without showing any signs of external leakage, or permanent deformation during or after the test.

3.4.1.3.2 External leakage - The hydraulic pump and stroke control shall show no signs of external leakage, during the tests.

3.4.1.3.3 No load power loss - The no load power loss of the hydraulic pump shall not exceed 125 inch-pounds, with the pump shaft driven at 1,750 to 1,800 revolutions per minute (rpm) and the slide block at neutral position.

3.4.1.3.4 Noise and vibration - The hydraulic pump and stroke control shall not evidence any excessive noise and vibration.

3.4.1.3.5 Internal leakage (dead spot) - The internal leakage of the hydraulic pump and stroke control shall meet the following requirements with the pump shaft rotating at 1,750 to 1,800 rpm and the high pressure ports of the pump blocked:

(a) The total displacement of the piston, Drawing 1676894, necessary to provide 100 psi differential at high pressure ports "A" and "B" shall not exceed 0.001 inch from the median of the neutral position.

(b) A pressure differential of 1,000 psi at ports "A" and "B" shall not require a displacement of the piston of more than 0.007 inch from the median of the neutral position.

(c) A pressure differential of 2,000 psi at ports "A" and "B" shall not require a displacement of the piston of more than 0.015 inch from the median of the neutral position.

3.4.1.3.6 Gear pump relief valves - The gear pump relief valves shall be set to deliver fluid at 100 ±10 psi for the supercharging pressure output and 260 ±15 psi for the control pressure output.

3.4.1.3.7 Gear pump delivery and leakage - The gear pump shall deliver a minimum of 1,100 cubic inches per minute at 100 ±10 psi from the supercharging

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

pressure output port, and a minimum of 980 cubic inches per minute at 260 ±15 psi at the control pressure output port. The pump shall be driven at 1,750 to 1,800 rpm.

The combined leakage from both pumps discharged from the main pump housing shall not exceed 120 cubic inches per minute.

3.4.1.3.8 High pressure relief valves - The high pressure relief valves shall be set to operate at 3,500 ±150 psi, with the pump stroke at 0.035 inches in either direction from neutral. Increasing the pump stroke to 0.316 inches in either direction shall cause a maximum of 800 psi rise in the high pressure over the initial relief valve operating pressure.

3.4.1.3.9 Combined leakage, high pressure relief valves and supercharge check valves - The combined leakage from the high pressure relief valves and supercharging check valves shall not exceed 16 cubic inches per minute, measured from the supercharge pressure output port when hydraulic fluid is supplied at 3,200 ±50 psi to either high pressure port of the hydraulic pump. The pump stroke is positioned at neutral, and the pump shaft is restrained.

3.4.1.3.10 Supercharge check valve operation - The supercharge check valves shall open at 12 ±4 psi fluid pressure.

3.4.1.3.11 Total leakage, except between flat valve openings - The total leakage, except between flat valve openings, shall not exceed 200 cubic inches per minute and the shaft rotation (if any), due to leakage, shall be uniform for each quadrant of full revolution under the following pump conditions:

(a) Slide block positioned at maximum deflection in either direction.

(b) Driving shaft free to rotate.

(c) Hydraulic fluid at a pressure of 3,200 ±50 psi applied to one high pressure port with the other high pressure port blocked.

(d) Leakage measured as discharged from the pump case.

3.4.1.3.12 Leakage between flat valve openings - The leakage between the flat valve openings shall not exceed 100 cubic inches per minute with the following pump conditions:

(a) Slide block positioned at neutral.

(b) Hydraulic fluid at a pressure of 3,200 ±50 psi applied to one high pressure port of the pump.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

(c) Leakage measured at the other high pressure port of the pump. The leakage shall not exceed the requirement for any of 26 equally spaced minimum seal restrained positions of the drive shaft rotation.

3.4.1.3.13 Dielectric strength and insulation resistance - The electric solenoid in the stroke control housing shall meet the requirements as defined in 3.3.

3.4.1.3.14 Solenoid valve operation - The valve spool, Drawing 1778642, shall move with no evidence of binding to the extreme of the stroke when the solenoid is energized with 103 ±0.5 volts alternating current (a.c.) and a hydraulic fluid pressure of 260 ±15 psi is applied to one port of the valve. When the solenoid is deenergized, the valve spool shall return to its normal position.

3.4.1.3.15 Combined leakage across the lands of valve spool and piston - The combined leakage across the lands of valve spool, Drawing 1778642, and piston, Drawing 1676894, shall not exceed 5 cubic inches per minute with the solenoid valve energized and hydraulic fluid applied at a pressure of 260 ±15 psi to each of the two valve ports in turn. Leakage shall be measured from the port opposite to that which the pressure is applied.

3.4.1.3.16 Combined leakage past the lands of control pistons and follow-up valve spool - The combined leakage past the lands of the control pistons, Drawings 1778662 and 1676907, and follow-up valve spool, Drawing 1683740, shall not exceed 75 cubic inches per minute with hydraulic fluid at a pressure of 260 ±15 psi applied to the input port for the control pistons and with leakage measured from the pump case.

3.4.1.3.17 Mechanical freedom of follow-up valve spool and piston - The break away force required to move the follow-up valve spool, Drawing 1683740, and the piston, Drawing 1676894, shall not exceed 8 ounces, with the centering spring, Drawing 1683770, removed.

3.4.1.3.18 Limit stop relief valve - The limit stop relief valves, Drawing 1284008, shall crack at 50 ±5 psi hydraulic fluid pressure. When the fluid pressure is increased to 90 psi, a minimum of 175 cubic inches per minute shall flow through the valve. The pressure at which the valve resets shall be 45 psi minimum.

3.4.1.3.19 Servo-valve - The servo-valve shall meet the requirements of Drawing 1284023.

3.4.1.4 Electric motors - The electric motors shall meet the requirements specified on the Drawings of LD's 286717 and 272632.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

3.4.2 Combined hydraulic pump and stroke control and the hydraulic B-end motor - The combined hydraulic pump and stroke control and the hydraulic motor, B-end, shall meet the following requirements.

3.4.2.1 Noise and vibration - The combined hydraulic pump and stroke control and the hydraulic motor, B-end, shall not evidence any excessive noise and vibration.

3.4.2.2 Pump control piston displacement related to hydraulic motor B-end no load output speed - With the hydraulic pump driven at 1,750 to 1,800 rpm, the pump control piston displacement shall be related to the hydraulic motor, B-end, no load output speed in the following manner:

(a) When the control piston is displaced in either direction from neutral 0.316 inch, the hydraulic motor no load output speed shall be in the range of 1,660 to 1,760 rpm.

(b) The neutral dead spot shall not exceed 0.002 inch.

(c) When the control piston is displaced from neutral in either direction, the no load output speed, as attained for the same piston displacements in either direction, shall not differ by more than 4 rpm, for piston displacements of 0.000 inch, 0.002 inch, 0.004 inch and 0.006 inch; 7.5 rpm for a piston displacement of 0.010 inch; 15 rpm for a piston displacement of 0.020 inch; and 25 rpm for a piston displacement of 0.150 inch and 0.316 inch.

3.4.2.3 Variation in output shaft speed - With the hydraulic pump driven at 1,750 to 1,800 rpm and the hydraulic motor output shaft loaded to produce a differential pressure at the motor high pressure ports of 1,000 ±50 psi, the maximum variations of the time for quarter turns of the output shaft at an intended speed of one rpm in either direction shall not exceed 10 percent.

3.4.2.4 Internal leakage -

(a) With the hydraulic pump driven at 1,750 to 1,800 rpm and the hydraulic motor shaft restrained to prevent rotation, the deflection of the stroke control piston, Drawing 1676894, necessary to produce 100 psi differential at high pressure ports "A" and "B" of the hydraulic motor, shall not exceed 0.001 inch from the median of the neutral position. In the same manner, the deflection to produce 1,000 psi differential at ports "A" and "B" shall not exceed 0.009 inch from the median of the neutral position. Also, in the same

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

manner, the deflection to produce 2,000 psi differential at ports "A" and "B" shall not exceed 0.019 inch from the median of the neutral position.

(b) With the hydraulic pump driven at 1,750 to 1,800 rpm, and the pump stroke displaced to cause a hydraulic motor shaft speed of approximately 40 rpm unloaded, apply a torque load to the hydraulic motor shaft to cause a fluid pressure differential to exist at the hydraulic motor high pressure ports. Under these conditions, the following results shall be attained.

(1) With the hydraulic motor loaded to 200 psi, the motor speed shall be 34 rpm, minimum, clockwise and counterclockwise.

(2) With the hydraulic motor loaded to 400 psi, the motor speed shall be 27 rpm, minimum, clockwise and counterclockwise.

(3) With the hydraulic motor loaded to 600 psi, the motor speed shall be 20 rpm, minimum, clockwise and counterclockwise.

(4) With the hydraulic motor loaded to 800 psi, the motor speed shall be 12 rpm, minimum, clockwise and counterclockwise.

(5) With the hydraulic motor loaded to 1,000 psi, the motor speed shall be 4 rpm, minimum, clockwise and counterclockwise.

3.4.2.5 Endurance test - With the combined pump and stroke control and the hydraulic motor, B-end, operating as a system, it shall operate for the periods of time indicated without any failure of parts, excessive wear, or unsatisfactory hydraulic system operation.

(a) For 500 hours under the following conditions:

(1) Hydraulic fluid, MIL-F-17111, be used and maintained at 120 degrees F to 180 degrees F.

(2) An oscillatory motion of 9/10 the full stroke each direction from neutral applied to the stroking mechanism at a period of 4 seconds total, approximating simple harmonic motion.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

d. First article sample subjected to the shock

(3) The output shaft loaded with an inertia wheel to cause 2,200 psi fluid pressure peaks to occur during the stroke.

(b) For 15 hours under the same conditions, except that the output shaft inertia shall be increased to cause 3,400 psi fluid pressure peaks during the stroke.

3.4.2.6 Power and efficiency test - The power and efficiency test shall be performed and the data specified below recorded. Hydraulic fluid, MIL-F-17111, shall be used and the fluid temperature maintained between 120 degrees F to 180 degrees F during the test. The hydraulic pump shall be driven at 1,750 to 1,800 rpm, and the slide block shall be positioned at 1/4, 1/2, 3/4, and full stroke each direction from neutral. The pressures on the high pressure lines shall be recorded at appropriate increments up to 2,250 psi. The following data shall be recorded:

(a) Hydraulic pump rpm.

(b) Hydraulic fluid temperature.

(c) Horsepower input to the hydraulic pump.

(d) Hydraulic line fluid pressures.

(e) Output torque of the hydraulic motor.

(f) Horsepower output from the hydraulic motor.

(g) Hydraulic motor rpm.

(h) Overall efficiency from hydraulic pump input to hydraulic motor output.

The torque output of the hydraulic motor shall be linear from a minimum of 1,950 pound-inches of torque at 2,500 psi line pressure to 0 pound-inches at the breakaway pressure of the B-end.

The minimum overall efficiency shall be as follows:

(a)	Full stroke	65 percent at 2,000 psi line pressure
	or	
	3/4 stroke	55 percent at 800 psi line pressure
(b)	1/2 stroke	60 percent at 2,000 psi line pressure
		50 percent at 800 psi line pressure

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

(c)	1/4 stroke	40 percent at 2,000 psi line pressure 40 percent at 800 psi line pressure

3.4.3 Receiver-regulator - The receiver-regulator shall meet the following requirements.

3.4.3.1 Electrical zero - The receiver-regulator shall be electrically zeroed.

3.4.3.2 Dielectric strength and insulation resistance - The electrical wiring and the electrical equipment, except capacitors, in the receiver-regulator shall withstand a potential of 720 volts a.c. sinusoidal for a period of one minute, applied between the electrical terminals and housing. After meeting the requirements for dielectric strength, the insulation resistance of all components at 500 volts d.c. shall be not less than three megohms as measured between the electrical terminals and housing.

3.4.3.3 Limit stops - The limit stops shall be assembled in strict compliance with the requirements of drawings of LD's 272601 and 272600.

3.4.3.4 Accuracy of correspondence - Any mechanical position of the one speed, or 36 speed receiver-regulator dials shall correspond to like positions of the one and 36 speed dials of a dummy director, when the director and receiver-regulator are electrically connected in the same manner for operation as in the power drive. The following maximum errors shall apply: 36 speed accuracy of correspondence 0.5 minute error, 1 speed accuracy of correspondence 20 minutes error.

3.4.4 Hydraulic tubing LD 272637 (Power Drive part) - The hydraulic tubing shall meet the following requirement, in addition to the requirement on the Naval Ordnance Systems Command drawings.

3.4.4.1 Cleaning - The hydraulic tubing shall be cleaned in strict compliance with MIL-STD-419, or Sketch 308777.

3.4.5 Amplifier Chassis, LD 272613 - The Amplifier Chassis shall meet all the requirements of the amplifier, train or elevation, test data, Drawing 1877515.

3.4.6 Complete power drive and amplifier performance requirements - Each complete power drive and amplifier shall meet the following requirements. Hereafter in paragraph 3.4.6, reference to the term power drive shall be interpreted to mean a power drive and the associated amplifier. The power drive shall be tested, utilizing the Dynamic Test Stand, Train, Drawing LD 272644 and Elevation, Drawing LD 272643.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

3.4.6.1 Electrical-mechanical zero feed back potentiometer - The feed back potentiometer operating shaft shall be centered in its travel, so that the d.c. voltage measured between the two wiper arms shall be 0 ±0.25 volt.

3.4.6.2 Mechanical-hydraulic balance, requirement - The mechanical-hydraulic balance, as indicated by the voltage, independent of oscillations, at the torque motor when a power drive is synchronized to a static signal, shall not exceed ±2.0 volts d.c. This requirement shall be met when the fluid temperature at the hydraulic pump case drain is in the range of 120° to 135° F.

3.4.6.3 Velocity requirements -

3.4.6.3.1 Lowest maximum velocity - The lowest maximum velocity in either direction shall be at least:

Train - 38° per second
Elevation - 24° per second

3.4.6.3.2 Maximum velocity - The power drive maximum velocities in either direction shall not exceed the following:

Train - 45° per second
Elevation - 26° per second

3.4.6.4 Acceleration and deceleration requirements -

3.4.6.4.1 Lowest maximum acceleration and deceleration - The following lowest maximum acceleration and deceleration in either direction shall be at least:

Train - 55° per second per second
Elevation - 85° per second per second

3.4.6.4.2 Maximum acceleration and deceleration - The maximum acceleration and deceleration in either direction shall not exceed the following:

Train - 150° per second per second
Elevation - 250° per second per second

3.4.6.5 Limit stop requirements -

3.4.6.5.1 Deceleration - The limit stops shall decelerate the power drives to rest within the acceleration limits specified in 3.4.6.4.2.

3.4.6.5.2 Limit stop position settings - The limit stop shall bring the power drive to rest within the following positions:

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

Receiver-Regulator Dial Readings

Right Train:	177° 10 minutes plus or minus 20 minutes
Left Train:	182° 50 minutes plus or minus 20 minutes
Elevation:	6,990 minutes plus or minus 20 minutes
Depression:	1,930 minutes plus or minus 20 minutes

3.4.6.6 Transfer valve, needle choke, valve spool timing - The transfer valve, valve spool shall return to the by-pass condition within 0.8 to 1.0 seconds time when the solenoid is de-energized.

3.4.6.7 Local control power drive requirements -

3.4.6.7.1 Local control safety switch - The local control safety switch, when released, shall de-energize the stroke control solenoid valve and the transfer valve solenoid.

3.4.6.7.2 Velocity of the power drives - The velocity of the power drives shall be proportional to any given deflection of the local control. With the handle deflected fully in either direction, the local control gain shall be capable of adjustment for power drive velocity of 10 degrees per second.

3.4.6.8 Accuracy requirements -

3.4.6.8.1 Static accuracy of correspondence - The static accuracy of correspondence of the power drives, not including the hunt, shall not exceed ±2 minutes error.

3.4.6.8.2 Hunt - The hunt of the power drive shall not exceed ±2.5 minutes.

3.4.6.8.3 Constant velocity errors - The constant velocity errors of the power drives shall not exceed the following:

Constant Velocity	Max. Allowable Error (Minutes)
Train	
1° to 2° per second	3
15° to 20° per second	7
30° to 35° per second	10
Elevation	
1° to 2° per second	3
10° to 15° per second	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

3.4.6.8.4 Simple harmonic motion - The simple harmonic motion errors of the power drives shall not exceed the following:

Simple Harmonic Motion	Max. Allowable Error (Minutes)
Train	
5° roll, 9 second period	6
30° roll, 9 second period	12
17° roll, 4.5 second period	25
Elevation	
5° roll, 9 second period	6
20° roll, 9 second period	10
5° roll, 4.5 second period	12

3.4.6.9 Synchronism firing interlock and synchronism indicator circuit requirements - The synchronism firing interlock circuit shall prevent firing and the synchronism indicator shall indicate when the error between the train or elevation power drive and director exceeds 20 minutes ±10 percent.

3.4.6.10 Synchronism - The power drives shall be considered synchronized or in synchronism during the synchronization tests when the error between power drive position and position order is 20 minutes or less. For Stow position, the power drive shall be considered synchronized for errors of 5 minutes or less.

3.4.6.11 Synchronization (stationary signals) - The power drives shall synchronize to the following stationary order signals in either direction from the synchronous position within the time listed:

Displacement Between Power Drive Position and Position Order	Maximum Time (Seconds) To Synchronize	
	Train	Elevation
170°	7.5	--
90°	6.0	--
60°	---	5
10°	3	3

3.4.6.12 Synchronization (return to Stow) and guide pin interlock relay operation, Elevation Power Drive - The Elevation Power Drive shall synchronize from the Operate to the Stow position within 5.0 seconds with not more than one

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

overshoot of a 30 minute maximum error when the positions are 60 degrees apart. When the power drive is positioned to Stow, the guide pin interlock circuit shall operate.

3.4.6.13 Synchronization (simple harmonic motion signals) - The power drives shall synchronize to simple harmonic motion order signals within the times specified in the following table. The power drives shall be displaced from the midpoint of the roll, decreasing direction, and the synchronizing action shall be initiated at the times indicated in the table.

Simple Harmonic Motion Order Signals and Power Drive Displacement	Time Synchronization Initiated - Point of Roll - Velocity and Direction	Maximum Time (Seconds) to Synchronize	
		Train	Elev.
15° amplitude 9.0 second	Zero Velocity, Increasing direction	7.0	6.0
Displaced 18° from Midpoint of Roll, Decreasing Direction	Maximum Velocity, Increasing direction	7.0	6.0
	Zero Velocity, Decreasing direction	7.0	6.0
	Maximum Velocity, Decreasing direction	7.0	6.0
5° amplitude, 4.5 second period	Zero Velocity, Increasing direction	6.0	6.0
Displaced 10° from Midpoint of Roll Decreasing Direction	Maximum Velocity, Increasing direction	6.0	6.0
	Zero Velocity, Decreasing direction	6.0	6.0
	Maximum Velocity, Decreasing direction	6.0	6.0

3.4.6.14 Power failure - When electrical power to the transfer valve and the stroke control solenoid is interrupted, a power drive shall decelerate and come to rest within the limits specified in 3.4.6.4.2.

3.4.6.15 Fluid pressure switch - The fluid pressure switch in the 100 psi supercharge fluid line shall operate as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

(a) Close within 3 seconds after the power drive motor reaches operating speed.

(b) Open when the power drive motor stops.

3.4.6.16 Hydraulic transmission dead spot - The maximum amount of stroke movement in both directions from the median of the neutral position, to produce the differential high pressures indicated, shall be as follows:

High Pressure Differential (psi)	Train (inches)	Elevation (inches)
100	.001	.0013
1,000	.010	.013
2,000	.020	.026

The ratio of the measurements recorded in either direction for the 1,000 psi and 2,000 psi tests shall not be greater than 2 to 1.

3.5 Environmental requirements - The power drives shall meet the environmental requirements of this specification during or after tests defined or specified in 4.5.

3.5.1 Temperature and humidity - The power drives shall meet the requirements of 3.4 when operating in ambient temperatures of 32 to 120 degrees Fahrenheit and in ambient relative humidities up to 95 percent. The power drives shall be capable of starting and operating satisfactorily in temperatures between 0° and 32° F and between 120° and 160° F, but not necessarily at specified rates and accuracies.

3.6 Clean-up - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.7 Workmanship - The equipment, including all parts, components, and the assembly, shall be constructed in a manner to assure compliance with all the requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges.

Source: https://assist.dla.mil -- Downloaded: 20260-19T18:29Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection -

4.2.1 Individual - Unless otherwise specified, each Power Drive, Train, Mk 61 Mod 2, Power Drive, Elevation, Mk 62 Mod 2, and Amplifier, Power Drive shall be subjected to the performance tests of 4.4 to assure the compliance with the requirements of 3.4.

4.2.2 Sampling - Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products shall be in accordance with the provisions set forth in MIL-STD-105. From each lot, select a random sample of the Power Drive, Train, Mk 61 Mod 2, Power Drive, Elevation, Mk 62 Mod 2, and Amplifier Power Drive in accordance with MIL-STD-105, Inspection Level S-2, Acceptance Number Zero, for submission to a Naval Ordnance Systems Command designated activity to determine compliance of the samples with the requirements of the contract, specifications and drawings. The units shall be considered satisfactory if they meet all specifications and pertinent drawing requirements when tested in accordance with 4.4 and 4.5. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot, or the cessation of production as determined by the procuring activity.

4.2.2.1 Inspection lot - Inspection lot definition, formation and size shall be in accordance with MIL-STD-105.

4.3 Inspection of preparation for delivery - Preparation for delivery of the Power Drives and Amplifier Chassis shall be inspected to determine conformance to Section 5 of this specification.

4.4 Performance tests - These tests are to be accomplished on each Power Drive, Train, Mk 61 Mod 2, Power Drive, Elevation, Mk 62 Mod 2, and Amplifier, Power Drive, being submitted for acceptance under contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

NOTE: Alternate performance test in accordance with manufacturer's test procedures may be used in lieu of 4.4.5.1.1, 4.4.5.1.2 and 4.4.5.2, provided they have approval by the procuring activity prior to use and ensure the quality of the units to the standards of the periodic production tested units.

4.4.1 Test equipment - The following items of test equipment are required to perform the performance tests set forth in this specification:

4.4.1.1 Standard measuring instruments -

Two (2) dial indicators (graduated in 0.001 inches range to 0.5 inches) accurate to 1 division.

One (1) spring scale (0 to 16 ounces, graduated in ounces) accuracy 5 percent full scale.

One (1) pneumatic pressure gage (range 0 to 25 psi, graduated in pounds per square inch), accuracy 1/2 of 1 percent of the scale range.

Two (2) hydraulic pressure gages (range 0 to 300 psi, graduated in pounds per square inch), accuracy 1/2 of 1 percent of the scale range.

Three (3) hydraulic pressure gages (range 0 to 5,000 psi, graduated in pounds per square inch), accuracy 1/2 of 1 percent of the scale range.

One (1) a.c. voltmeter, rectifier type (range 0 to 150 volts, graduated in 0.5 volts, 1,000 ohms per volt), accuracy 3 percent full scale.

Two (2) mercury bulb thermometers (graduated in degrees Fahrenheit, range 0 to 212 degrees), accuracy plus or minus 1 degree.

Two (2) measuring flasks (range 0-50 cubic inches, graduated in 0.5 cubic inches).

Two (2) hydraulic pressure gages (range 0 to 60 psi, graduated in pounds per square inch), accuracy 1/2 of 1 percent of the scale range.

Two (2) measuring flasks (range 0 to 250 cubic inches, graduated in cubic inches).

One (1) measuring flask (range 0 to 10 cubic inches, graduated in 0.5 cubic inches).

Two (2) tachometers (range 0 to 2,000 rpm, graduated in revolutions per minute), accuracy 5 percent full scale.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T18:29Z
Check the source to verify that this is the currentversion before use.

One (1) stop watch (graduated in 0.01 seconds).

One (1) d.c. voltmeter (range 0-10, 0-100 volts, graduated in 0.5 volts, 1,000 ohms per volt), accuracy 3 percent full scale.

One (1) a.c. voltmeter, electronic type (range 10 millivolts to 200 volts) full scale, accuracy 3 percent full scale.

One (1) d.c. voltmeter (range 1 volt to 200 volts full scale, 20,000 ohms per volt), accuracy 3 percent full scale.

One (1) oscilloscope, 5 inch cathode ray tube (range, peak to peak voltage 200 volts). Dumont Model 304A or equal.

One (1) ohmmeter, Multirange type (range 0 to 10 megohms), accuracy 5 percent full scale.

Two (2) dial indicators (graduated in 0.001, range to 0.015), accurate to 1 division.

One (1) torque wrench (0 to 40 in. oz. range), accuracy within 10 percent full scale.

4.4.1.2 Special test equipment -

One (1) Hi-pot Tester (1,500 VAC) for dielectric strength testing, full scale accuracy 5 percent.

One (1) Megohmmeter (500 volts d.c. and 250 volts d.c.) for insulation resistance testing, full scale accuracy 3 percent.

Two (2) Dummy Directors Mk 3 Mod 8 and Error Recorder Mk 12, LD 272569.

One (1) Dynamometer Test Stand, for testing hydraulic pump, stroke control and hydraulic motor, B-end. (See 1/).

One (1) Hydraulic Fluid Supply (range 0 to 5,000 psi, 0.5 gallon per minute) for hydrostatic pressure testing. (See 1/).

One (1) Compressed Air Supply (range 0 to 10 psi) for hydrostatic pressure testing.

One (1) Hydraulic Test Stand, for testing sub-assemblies of hydraulic equipment. (See 1/).

One (1) Dynamic Test Stand, Train Power Drive Mk 61 Mod 2, LD 272644.

One (1) Dynamic Test Stand, Elevation Power Drive Mk 62 Mod 2, LD 272643.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

One (1) Test Panel, Assembly, Amplifier, Drawing 1478765.

1/: These special test equipments are used in testing components which are procured under a licensing agreement. General description of equipment is listed for information.

4.4.2 Acceptance procedures -

4.4.2.1 Performance tests - The supplier shall conduct the performance tests as specified in 4.2.1 to assure that Power Drive, Train, Mk 61 Mod 2, Power Drive, Elevation, Mk 62 Mod 2, and Amplifier, Power Drive, is in compliance with the requirements of 3.2, 3.4, 3.6 and 3.7.

4.4.3 Test conditions - Unless otherwise specified the Power Drive, Train, Mk 61 Mod 2, Power Drive, Elevation, Mk 62 Mod 2 and Amplifier, Power Drive, shall be subjected to performance tests under the following conditions:

4.4.3.1 Temperatures - Room ambient 32 degrees F to 120 degrees F.

4.4.3.2 Altitude - Normal ground atmospheric pressure.

4.4.3.3 Humidity - Room ambient to 95 percent relative maximum.

4.4.4 Test and inspection equipment and facilities - The manufacturer shall furnish and maintain all necessary test equipment and facilities and personnel for conducting all performance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit the conductance of the required performance tests.

4.4.5 Test procedure - The following tests shall be performed, using the procedures outlined. Test data shall be recorded and tabulated for all tests. When applicable, Factory Acceptance Test, NAVORD OD 14430 (Second Revision), shall be used to record and tabulate the results of the tests.

4.4.5.1 Power unit - When assembled in accordance with Drawings 2319716 and 2319715, and tested, the watertightness requirements, hydrostatic requirements and gearing backlash requirements shall not exceed the values specified on the drawings.

4.4.5.1.1 Hydraulic motor and transfer valve - The hydraulic motor and transfer valve shall be tested for performance in accordance with, and meet the requirements of Drawing 1691753.

4.4.5.1.2 Hydraulic pump and stroke control -

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

4.4.5.1.2.1 Internal hydrostatic pressure - Apply a hydrostatic fluid pressure of 4,550 ±50 psi internally to the hydraulic pump and stroke control, except the pump case and the stroke control servo-valve housing. Outlet ports may be blocked to achieve this test. The hydraulic pump and stroke control shall meet the requirements of 3.4.1.3.1. The pump case and stroke control servo-valve housing shall be pressurized to 5 psi hydrostatically and shall meet the requirements of 3.4.1.3.1.

4.4.5.1.2.2 External leakage - During operation, the hydraulic pump and stroke control shall not exceed the external leakage requirement of 3.4.1.3.2.

4.4.5.1.2.3 No load power loss - The no load power loss of the hydraulic pump when driven at 1,750 to 1,800 rpm, as measured at the driving source, shall meet the requirements of 3.4.1.3.3.

4.4.5.1.2.4 Noise and vibration - During operational tests, the hydraulic pump shall meet the noise and vibration requirement of 3.4.1.3.4.

4.4.5.1.2.5 Internal leakage (dead spot) - The hydraulic pump shall be driven at 1,750 to 1,800 rpm with the high pressure ports blocked. The internal leakage (dead spot) shall be within the requirements of 3.4.1.3.5.

4.4.5.1.2.6 Gear pump relief valves - The gear pump relief valves shall be set so that the pump will deliver hydraulic fluid under the pressures of 3.4.1.3.6.

4.4.5.1.2.7 Gear pump delivery and leakage - The pump shall be driven at 1,750 to 1,800 rpm. The gear pump shall be capable of delivering hydraulic fluid at the rate of flow and meet the leakage requirements of 3.4.1.3.7.

4.4.5.1.2.8 High pressure relief valves - The hydraulic pump and stroke control shall be operated with the high pressure ports blocked. The relief valve settings shall meet the requirements of 3.4.1.3.8.

4.4.5.1.2.9 Combined leakage, high pressure relief valves and supercharge check valves - Hydraulic fluid shall be applied at a pressure of 3,200 ±50 psi to either high pressure port of the pump with the slide block at neutral and the pump shaft restrained. The combined leakage of the high pressure relief valves and the supercharging check valves, measured at the supercharge pressure output port, shall not exceed the requirements of 3.4.1.3.9.

4.4.5.1.2.10 Supercharge check valve operation - The supercharge check valve shall be tested with a hydraulic fluid pressure and shall open within the requirements of 3.4.1.3.10.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

4.4.5.1.2.11 Total leakage, except between flat valve openings - Hydraulic fluid at a pressure of 3,200 ±50 psi shall be applied to one high pressure port of the pump with the other high pressure port blocked. The slide block shall be positioned and restrained at maximum deflection in either direction. The driving shaft shall be free to rotate. Under these conditions, the leakage measured from the pump case discharge shall meet the total leakage requirement, except between flat valve openings of 3.4.1.3.11.

4.4.5.1.2.12 Leakage between flat valve openings - Hydraulic fluid at a pressure of 3,200 ±50 psi shall be applied to one high pressure port of the pump. The slide block shall be positioned and restrained at neutral. Rotating the input shaft to 26 equally spaced minimum seal positions and restraining the shaft at each position, the leakage measured from the other high pressure port shall meet the leakage between flat valve openings requirement of 3.4.1.3.12.

4.4.5.1.2.13 Dielectric strength and insulation resistance - The electrical solenoid in the stroke control housing shall be tested for dielectric strength and insulation resistance with the external circuit disconnected. The solenoid shall meet the requirements of 3.4.1.3.13.

4.4.5.1.2.14 Solenoid valve operation - Hydraulic fluid at a pressure of 260 ±15 psi shall be applied to one port of the solenoid valve and the solenoid energized and de-energized. The valve spool, Drawing 1778642, shall operate within the requirements of 3.4.1.3.14.

4.4.5.1.2.15 Combined leakage across the lands of valve spool and piston - Hydraulic fluid shall be applied at a pressure of 260 ±15 psi to each of the valve ports of the solenoid valve in turn with the solenoid energized. The leakage measured at the port opposite to which the fluid pressure is applied shall not exceed the requirements of 3.4.1.3.15.

4.4.5.1.2.16 Combined leakage past the lands of control pistons and follow-up valve spool - Hydraulic fluid shall be applied at a pressure of 260 ±15 psi to the input port of the control pistons of the stroke control. The combined leakage past the lands of the control pistons and follow-up valve spool measured at the pump case discharge shall not exceed the requirement of 3.4.1.3.16.

4.4.5.1.2.17 Mechanical freedom of follow-up valve spool and piston - With a spring scale attached to the follow-up valve spool and piston, exerting a force on the scale, the follow-up valve and piston shall move within the requirements of 3.4.1.3.17.

4.4.5.1.2.18 Limit stop relief valve - Hydraulic fluid shall be applied to the ports of the relief valve, in turn. The limit stop relief valves shall meet the requirement of 3.4.1.3.18.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T18:29Z
Check the source to verify that this is the current version before use.

4.4.5.1.2.19 Servo-valve - The torque motor servo-valve shall be tested for performance in accordance with, and meet the requirements of, Drawing 1284023.

4.4.5.1.3 Electric motors - The electric motors shall be assembled and tested in accordance with LD's 286717 and 272632. The requirements of the drawings shall be met.

4.4.5.2 Combined hydraulic pump and stroke control and the hydraulic B-end motor -

4.4.5.2.1 Noise and vibration - During operational tests, the noise and vibration of the combined hydraulic pump and stroke control and hydraulic motor shall meet the requirement of 3.4.2.1.

4.4.5.2.2 Pump control piston displacement - The hydraulic pump shall be driven at 1,750 to 1,800 rpm. The speed of the output shaft of the hydraulic motor shall be measured without load. The displacement of the control piston of the pump and the output speed of the hydraulic motor shall meet the requirements of 3.4.2.2.

4.4.5.2.3 Variations in output shaft speed - The hydraulic pump shall be driven at 1,750 to 1,800 rpm and the shaft of the hydraulic motor loaded to produce a differential pressure at the motor high pressure ports of 1,000 ±50 psi at an intended speed of one rpm in either direction. The variations in the output shaft speed shall be within the requirement of 3.4.2.3.

4.4.5.2.4 Internal leakage - The hydraulic pump shall be driven at 1,750 to 1,800 rpm and the fluid pressure measured in the high pressure lines connecting the hydraulic pump to the hydraulic motor. The pump stroke shall be positioned, and the output shaft of the hydraulic motor loaded in accordance with the requirements of 3.4.2.4. The deflection of the pump stroke, the fluid pressure in the high pressure lines and the output speed of the hydraulic motor shall meet the requirements of 3.4.2.4.

4.4.5.3 Receiver-regulator -

4.4.5.3.1 Electrical zero - The receiver-regulator synchro units shall be electrically zeroed in accordance with the requirements of 3.4.3.1.

4.4.5.3.2 Dielectric strength and insulation resistance - The receiver-regulator electrical wiring and electrical components with external circuits disconnected shall be subjected to dielectric strength and insulation resistance tests and meet the requirements of 3.4.3.2.

4.4.5.3.3 Limit stops - The receiver-regulator limit stops shall be assembled, installed and positioned in strict compliance with the requirements of 3.4.3.3.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

4.4.5.3.4 Accuracy of correspondence - The receiver-regulator shall be electrically connected to a dummy director in the same manner as for operation in the power drive. Mechanical positions of the receiver-regulator dials shall correspond to like positions of the director dials, when electrically aligned, within the requirements of 3.4.3.4.

4.4.5.4 Hydraulic tubing -

4.4.5.4.1 Cleaning - Cleaning of all hydraulic tubing shall be in strict compliance with requirements of 3.4.4.1.

4.4.5.5 Amplifier chassis - The amplifier chassis shall be tested in accordance with and meet the requirements of 3.4.5.

4.4.5.6 Complete power drive and amplifier -

4.4.5.6.1 Electrical-mechanical zero, feed back potentiometer - A d.c. voltmeter (1,000 ohms per volt) shall be applied to terminal board TB1-19 and TB1-25 of the receiver-regulator. The feed back potentiometer operating shaft adjusted for neutral position of the stroke control shall meet the requirements of 3.4.6.1.

4.4.5.6.2 Mechanical-hydraulic balance - A d.c. voltmeter (1,000 ohms per volt) shall be applied to the amplifier terminal board TB1-6 and TB1-8. The maximum voltage independent of oscillations when the power drive is synchronized shall meet the requirements of 3.4.6.2.

4.4.5.6.3 Velocity -

4.4.5.6.3.1 Lowest maximum velocity - The power drive shall be tested to determine that it meets the requirement for lowest maximum velocity of 3.4.6.3.1.

4.4.5.6.3.2 Maximum velocity - The power drive shall be tested to determine that it meets the requirement for maximum velocity of 3.4.6.3.2.

4.4.5.6.4 Acceleration and deceleration -

4.4.5.6.4.1 Lowest maximum acceleration and deceleration - The power drive shall be tested to determine that it meets the requirement for lowest maximum acceleration of 3.4.6.4.1.

4.4.5.6.4.2 Maximum acceleration and deceleration - The power drive shall be tested to determine that it meets the requirements for maximum acceleration of 3.4.6.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

4.4.5.6.5 Limit stop -

4.4.5.6.5.1 Deceleration - During operation of the power drive, at any velocity, the limit stops shall not allow the power drive to exceed the deceleration requirements of 3.4.6.5.1.

4.4.5.6.5.2 Limit stop position settings - During operation of the power drive, at any velocity, the limit stops shall not allow the power drive to exceed the requirements of 3.4.6.5.2.

4.4.5.6.6 Transfer valve, needle choke, valve spool timing - During operation of the power drive, when the stroke control is deflected, with the stroke control solenoid de-energized, the transfer valve, valve spool shall return within the requirements of 3.4.6.6 when its solenoid is de-energized.

4.4.5.6.7 Local control power drive -

4.4.5.6.7.1 Local control safety switch - Releasing the local control safety switch shall cause it to meet the requirements of 3.4.6.7.1.

4.4.5.6.7.2 Velocity of the power drives - When the handle of the local control is deflected fully in either direction, the power drive shall attain the velocity in accordance with 3.4.6.7.2.

4.4.5.6.8 Accuracy -

4.4.5.6.8.1 Static accuracy of correspondence - The power drive when synchronized to the dummy director shall meet the requirement for static accuracy in 3.4.6.8.1.

4.4.5.6.8.2 Hunt - The power drive when synchronized to the dummy director shall meet the requirement for hunt in 3.4.6.8.2.

4.4.5.6.8.3 Constant velocity errors - The power drive when synchronized to the dummy director and operated from constant velocity signals shall meet the requirements for constant velocity errors in 3.4.6.8.3.

4.4.5.6.8.4 Simple harmonic motion - The power drive when synchronized to the dummy director and operated from simple harmonic motion signals shall meet the requirements for simple harmonic motion in 3.4.6.8.4.

4.4.5.6.9 Synchronism firing interlock and synchronism indicator circuit - The power drive when out of synchronism with the dummy director shall cause the synchronism firing interlock and the synchronism indicator circuits to operate in accordance with the requirements in 3.4.6.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

4.4.5.6.10 Synchronism - The power drive shall be considered synchronized with the dummy director during the synchronization tests, when the error between the power drive and the director meet the requirements of 3.4.6.10.

4.4.5.6.11 Synchronization (stationary signals) - The power drive shall be tested to determine that it will synchronize to the fixed position order signals from the dummy director within the requirements of synchronization (stationary signals), in 3.4.6.11.

4.4.5.6.12 Synchronization (return to stow) and guide pin interlock relay operation - The power drive (elevation) shall be tested to determine that it will synchronize to the Stow order signals and the guide pin interlock circuit shall operate within the requirements of synchronization (return to Stow signals) and guide pin interlock relay operation in 3.4.6.12.

4.4.5.6.13 Synchronization (simple harmonic motion signals) - The power drive shall be tested to determine that it will synchronize to simple harmonic motion order signals from the director within the requirements of synchronization (simple harmonic motion signals) in 3.4.6.13.

4.4.5.6.14 Power failure - The power drive shall decelerate and come to rest when the stroke control solenoid and transfer valve solenoid power is removed, within the requirements for power failure in 3.4.6.14.

4.4.5.6.15 Fluid pressure switch - The power drive shall be tested to determine that the fluid pressure switch will operate within the requirements of 3.4.6.15.

4.4.5.6.16 Hydraulic transmission dead spot - The power drive shall be tested with the "B" end motor shaft locked to determine that the hydraulic transmission dead spot is within the limits specified in 3.4.6.16.

4.5 Periodic production tests - Periodic production tests shall consist of all of the tests of 4.4 and tests of 4.5.1 and 4.5.2.

4.5.1 Power and efficiency test - The combined hydraulic pump and stroke control and the hydraulic motor, B-end, shall be operated as a system. The pump shall be driven at 1,750 to 1,800 rpm and the slide block positioned at 1/4 stroke, 1/2 stroke, 3/4 stroke, and full stroke either direction from neutral. Hydraulic fluid MIL-F-17111 shall be used and maintained at a temperature of 120° F to 180° F. Data shall be recorded for the requirements of 3.4.2.6 and the test results shall meet the minimum requirements of 3.4.2.6.

4.5.2 Endurance test - The combined hydraulic pump and stroke control and the hydraulic motor, B-end, shall be operated as a system. The pump shall

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

be driven at 1,750 to 1,800 rpm. The stroke shall be deflected nine-tenths full stroke either direction from neutral at approximately simple harmonic, by a device connected to it. Hydraulic fluid MIL-F-17111 shall be used and maintained at 120° F to 180° F. The output shaft of the hydraulic motor shall be loaded with the inertia as required in 3.4.2.5. Time of operation, hydraulic line pressure, wear and failure shall meet the requirements of 3.4.2.5.

5. PREPARATION FOR DELIVERY

5.1 Power Drive, Train, Mk 61 Mod 2, Power Drive, Elevation, Mk 62 Mod 2 and Amplifier, LD 272613 shall be preserved, packaged, packed and marked in accordance with Levels A or C of OD 18388. (See 6.2.)

6. NOTES

6.1 Intended use - The Power Drives, Train, Mk 61 Mod 2 and Elevation, Mk 62 Mod 2, and the Amplifier are intended to be used to provide the motive power to position Launcher Mk 112 Mod 2.

6.2 Ordering data - Procurement documents should specify the following information with any options desired by the procuring activity:

(a) Title, number and date of this specification.

(b) Level of preservation required.

(c) Tools, special test equipment, spare parts and accessories to be furnished.

(d) Special marking required.

(e) Shipping information.

6.3 Storage surveillance - In the event the equipment specified herein is not installed within the period indicated for the prescribed level of preservation, the equipment shall be inspected for condition and necessary represervation shall be accomplished in accordance with procedures referenced in Section 5 of this document.

6.4 Description, operation and maintenance - For description and maintenance of Train Power Drive Mk 61, Elevation Power Drive Mk 62, and Amplifier, refer to OP 2385, Volume 2.

6.5 Definitions -

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

6.5.1 Supplier - Any organization under contract, capable of furnishing, or causing to be furnished, the product which meets the requirements of this specification.

This document has been issued as a companion document to MIL-L-23956A(OS).

Project No. 1055-N019

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the currentversion before use.

INDEX

CONTENTS	PARAGRAPH	PAGE
Acceleration and deceleration requirements	3.4.6.4, 4.4.5.6.4	14, 26
Acceptance procedures	4.4.2	22
Accuracy of correspondence	3.4.3.4, 4.4.5.3.4	13, 26
Accuracy requirements	3.4.6.8, 4.4.5.6.8	15, 27
Altitude	4.4.3.2	22
Amplifier chassis, LD 272613	3.4.5, 4.4.5.5	13, 26
APPLICABLE DOCUMENTS	2	1
Assembly requirements	3.4.1.1	6
Cleaning	3.4.4.1, 4.4.5.4.1	13, 26
Clean-up	3.6	18
Combined hydraulic pump and stroke control and the hydraulic B-end motor	3.4.2, 4.4.5.2	10, 25
Combined leakage across the lands of valve spool and piston	3.4.1.3.15, 4.4.5.1.2.15	9, 24
Combined leakage, high pressure relief valves and supercharge check valves	3.4.1.3.9, 4.4.5.1.2.9	8, 23
Combined leakage past the lands of control pistons and follow-up valve spool	3.4.1.3.16, 4.4.5.1.2.16	9, 24
Complete power drive and amplifier	4.4.5.6	26
Complete power drive and amplifier performance requirements	3.4.6	13
Conflicting requirements	3.2.9	6
Constant velocity errors	3.4.6.8.3, 4.4.5.6.8.3	15, 27
Deceleration	3.4.6.5.1, 4.4.5.6.5.1	14, 27
Definitions	6.5	29
Description	3.1	4
Description, operation and maintenance	6.4	29
Dielectric strength and insulation resistance	3.3.1, 3.4.1.3.13, 3.4.3.2, 4.4.5.1.2.13, 4.4.5.3.2	6, 9, 13 24, 25
Electric motors	3.4.1.4, 4.4.5.1.3	9, 25
Electrical-mechanical zero, feedback potentiometer	3.4.6.1, 4.4.5.6.1	14, 26
Electrical and hydraulic requirements	3.3	6
Electrical zero	3.4.3.1, 4.4.5.3.1	13, 25
Endurance test	3.4.2.5, 4.5.2	11, 28
Environmental requirements	3.5	18

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

CONTENTS	PARAGRAPH	PAGE
External leakage	3.4.1.3.2, 4.4.5.1.2.2	7, 23
Finishes, coatings, construction and dimensions	3.2.2	4
Fluid pressure switch	3.4.6.15, 4.4.5.6.15	17, 28
Gages	3.2.6	5
Gear pump delivery and leakage	3.4.1.3.7, 4.4.5.1.2.7	7, 23
Gear pump relief valves	3.4.1.3.6, 4.4.5.1.2.6	7, 23
General requirements	3.2	4
High pressure relief valves	3.4.1.3.8, 4.4.5.1.2.8	8, 23
Humidity	4.4.3.3	22
Hunt	3.4.6.8.2, 4.4.5.6.8.2	15, 27
Hydraulic fluid and hydraulic fluid temperatures	3.3.2	6
Hydraulic motor and transfer valve	3.4.1.2, 4.4.5.1.1	6, 22
Hydraulic pump and stroke control LD's 272633 and 272634	3.4.1.3, 4.4.5.1.2	7, 22
Hydraulic transmission dead spot	3.4.6.16, 4.4.5.6.16	18, 28
Hydraulic tubing, LD 272637 (Power Drive part)	3.4.4, 4.4.5.4	13, 26
Individual	4.2.1	19
Inspection lot	4.2.2.1	19
Inspection of preparation for delivery	4.3	19
Intended use	6.1	29
Interchangeability	3.2.5	5
Internal hydrostatic pressure	3.4.1.3.1, 4.4.5.1.2.1	7, 23
Internal leakage	3.4.2.4, 4.4.5.2.4	10, 25
Internal leakage (dead spot)	3.4.1.3.5, 4.4.5.1.2.5	7, 23
Leakage between flat valve openings	3.4.1.3.12, 4.4.5.1.2.12	8, 24
Limit stop position settings	3.4.6.5.2, 4.4.5.6.5.2	14, 27

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

CONTENTS	PARAGRAPH	PAGE
Limit stop relief valve	3.4.1.3.18, 4.4.5.1.2.18	9, 24
Limit stop requirements	3.4.6.5, 4.4.5.6.5	14, 27
Limit stops	3.4.3.3, 4.4.5.3.3, 4.4.5.6.5	13, 25, 27
Local control power drive requirements	3.4.6.7, 4.4.5.6.7	15, 27
Local control safety switch	3.4.6.7.1, 4.4.5.6.7.1	15, 27
Lowest maximum acceleration and deceleration	3.4.6.4.1, 4.4.5.6.4.1	14, 26
Lowest maximum velocity	3.4.6.3.1, 4.4.5.6.3.1	14, 26
Marking	3.2.4	5
Materials	3.2.1	4
Maximum acceleration and deceleration	3.4.6.4.2, 4.4.5.6.4.2	14, 26
Maximum velocity	3.4.6.3.2, 4.4.5.6.3.2	14, 26
Mechanical freedom of follow-up valve spool and piston	3.4.1.3.17, 4.4.5.1.2.17	9, 24
Mechanical-hydraulic balance, requirement	3.4.6.2, 4.4.5.6.2	14, 26
Moisture, fungus and corrosion	3.2.3	5
Name plate weight	3.2.4.1	5
No load power loss	3.4.1.3.3, 4.4.5.1.2.3	7, 23
Noise and vibration	3.4.1.3.4, 3.4.2.1, 4.4.5.1.2.4, 4.4.5.2.1	7, 10, 23, 25
NOTES	6	29
Ordering data	6.2	29
Other publications	2.2	3
Performance requirements and product characteristics	3.4	6
Performance tests	4.4, 4.4.2.1	19, 22
Periodic production tests	4.5	28
Power and efficiency test	3.4.2.6, 4.5.1	12, 28
Power failure	3.4.6.14, 4.4.5.6.14	17, 28
Power unit	3.4.1, 4.4.5.1	6, 22
PREPARATION FOR DELIVERY	5	29
Pump control piston displacement	3.4.2.2, 4.4.5.2.2	10, 25

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

CONTENTS	PARAGRAPH	PAGE
QUALITY ASSURANCE PROVISIONS	4	19
Quality conformance inspection	4.2	19
Receiver-regulator	3.4.3, 4.4.5.3	13, 25
Requests for deviations	3.2.8	6
REQUIREMENTS	3	4
Responsibility for inspection	4.1	19
Sampling	4.2.2	19
SCOPE	1	1
Scope	1.1	1
Servo-valve	3.4.1.3.19, 4.4.5.1.2.19	9, 25
Simple harmonic motion	3.4.6.8.4, 4.4.5.6.8.4	16, 27
Solenoid valve operation	3.4.1.3.14, 4.4.5.1.2.14	9, 24
Special test equipment	4.4.1.2	21
Standard measuring instruments	4.4.1.1	20
Static accuracy of correspondence	3.4.6.8.1, 4.4.5.6.8.1	15, 27
Storage surveillance	6.3	29
Supercharge check valve operation	3.4.1.3.10, 4.4.5.1.2.10	8, 23
Supplier	6.5.1	30
Synchronism	3.4.6.10, 4.4.5.6.10	16, 28
Synchronism firing interlock and synchronism indicator circuit requirements	3.4.6.9, 4.4.5.6.9	16, 27
Synchronization (return to Stow) and guide pin interlock relay operation, Elevation Power Drive	3.4.6.12, 4.4.5.6.12	16, 28
Synchronization (simple harmonic motion signals)	3.4.6.13, 4.4.5.6.13	17, 28
Synchronization (stationary signals)	3.4.6.11, 4.4.5.6.11	16, 28
Temperatures	4.4.3.1	22
Temperature and humidity	3.5.1	18
Test and inspection equipment and facilities	4.4.4	22
Test conditions	4.4.3	22

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

INDEX (Continued)

CONTENTS	PARAGRAPH	PAGE
Test equipment	4.4.1	20
Test procedure	4.4.5	22
Threads	3.2.7	5
Total leakage, except between flat valve openings	3.4.1.3.11, 4.4.5.1.2.11	8, 24
Transfer valve, needle choke, valve spool timing	3.4.6.6, 4.4.5.6.6	15, 27
Variation in output shaft speed	3.4.2.3, 4.4.5.2.3	10, 25
Velocity of the power drives	3.4.6.7.2, 4.4.5.6.7.2	15, 27
Velocity requirements	3.4.6.3, 4.4.5.6.3	14, 26
Workmanship	3.7	18

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:29Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION	
MIL-P-23957A(OS)	POWER DRIVE, TRAIN, MK 61 MOD 2 POWER DRIVE, ELEVATION, MK 62 MOD 2 CHASSIS, AMPLIFIER, LD 272613

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Weapons Services Office
Attention: Code SSS
Philadelphia, Pennsylvania 19112

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

Commanding Officer
Naval Weapons Services Office
Attention: Code SSS
Philadelphia, Pennsylvania 19112

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:29Z
Check the source to verify that this is the current version before use.